

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think>
He drives for 3 hours at 60 mph so he is $3 \times 60 = 180$ miles away from home
He spends 2 hours in standstill traffic so he doesn't move during that time
He spends next half hour driving at 30 mph so he covers $30 / 2 = 15$ miles
He has $4 - 2 - 0.5 = 1.5$ hours left to drive
He drives 1.5 hours at 80 mph so he covers $1.5 \times 80 = 120$ miles
Altogether he covers $15 + 120 = 135$ miles
Since he was 180 miles away and now he's 135 miles closer that means he is $180 - 135 = 45$ miles away from home
</think>
<answer> 45 </answer>

